

Test Results: Pretrained and fine-tuned Transformer

Overall Metrics

Metric	Value
Test loss	2.013650
Accuracy	44.72%
Macro precision	0.2088
Macro recall	0.2194
Macro F1-score	0.1974
Weighted precision	0.3697
Weighted recall	0.4472
Weighted F1-score	0.3867

Classification Report

Class	Precision	Recall	F1-score	Support
Bass	0.0000	0.0000	0.0000	0
Brass	0.0000	0.0000	0.0000	42
Chromatic Percussion	0.0000	0.0000	0.0000	17
Drums	0.7013	1.0000	0.8244	54
Guitar	0.5000	0.7129	0.5878	101
Organ	0.2903	0.3333	0.3103	27
Piano	0.6250	0.1515	0.2439	33
Pipe	0.1765	0.0882	0.1176	34
Reed	0.0000	0.0000	0.0000	9
Strings	0.0000	0.0000	0.0000	0
Strings (continued)	0.4211	0.5657	0.4828	99

Class	Precision	Recall	F1-score	Support
Synth Lead	0.0000	0.0000	0.0000	0
Synth Pad	0.0000	0.0000	0.0000	29
Macro average	0.2088	0.2194	0.1974	445
Weighted average	0.3697	0.4472	0.3867	445

Confusion Matrix

Rows represent true classes. Columns represent predicted classes.

True / Pred.	0	1	2	3	4	5	6	7	8	9	10	11	12
0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	2	3	0	2	0	0	32	2	0
2	0	0	0	17	0	0	0	0	0	0	0	0	0
3	0	0	0	54	0	0	0	0	0	0	0	0	0
4	5	0	0	3	72	14	2	1	0	0	4	0	0
5	0	0	0	0	2	9	0	11	0	0	5	0	0
6	15	0	0	0	12	1	5	0	0	0	0	0	0
7	0	0	0	2	27	0	0	3	0	0	2	0	0
8	0	0	0	0	1	0	0	0	0	0	8	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0	0
10	13	0	0	0	28	2	0	0	0	0	56	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	2	1	0	0	0	26	0	0

Class Index Key

Index	Class
0	Bass
1	Brass
2	Chromatic Percussion
3	Drums
4	Guitar
5	Organ
6	Piano
7	Pipe

Index	Class
8	Reed
9	Strings
10	Strings (continued)
11	Synth Lead
12	Synth Pad